AICTE VERIFIED STUDENT REGISTER

Problem Statement and Product Scope

What problem the prototype solves, who uses it, what is included, and what must not be claimed.

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Audience: AI coding agent + project owner

Document-control note
This is an implementation specification for a synthetic-data prototype, not an official AICTE standard, production approval, legal opinion, or proof of live government integration.

Source discipline: requirements from the PS 26263 implementation report and Sigma Batch curriculum are distinguished from proposed engineering choices. Any item that depends on a real policy, approval, API, credential trust chain, or deployment environment must be confirmed rather than guessed.

1. Problem statement

Problem statement (project-level summary)
AICTE needs a way to move from aggregate admission counts to student-level admission records that can be validated, institutionally certified, monitored, and selectively verified. Incoming college data may contain formatting errors, duplicate rows, seat-category overflow, inconsistent fields, or conflicting active enrolments. The proposed platform adds an AICTE-side verified register and oversight workflow; it does not recreate APAAR, ABC or DigiLocker.

Basis: PS 26263 report, pages 1-2. This wording is a plain-language project summary, not a claim that AICTE has approved this exact implementation.

2. Product goals

- Collect admission records from college CSV files; ERP/API input is a future extension unless explicitly implemented.
- Stage incoming records so that an upload is not automatically treated as the official register.
- Apply deterministic checks and report row-level errors, warnings and pending checks.
- Track course/year/category seat occupancy using configurable rules and explicit approved-capacity data.
- Require an authorised institution head to certify before a batch is committed.
- Keep enrolment lifecycle history and flag possible cross-college conflicts for review.
- Provide role-scoped dashboards and explainable, non-punitive risk signals.
- Demonstrate student-consent verification, QR status checking and tamper-evident audit history with synthetic data.

3. Users and high-level responsibilities

Role	Main responsibility	Boundary
College Operator / Maker	Upload files, fix validation issues, track submission status.	Cannot certify or submit the final commit.
College Head / Checker	Review submission, handle warnings/conflicts and certify.	Must belong to the institution being certified.
AICTE National Officer	National monitoring, case review, escalation and reporting.	Read/decision privileges only as explicitly defined; no implicit system-admin authority.
AICTE Regional Officer	Monitor and review cases in assigned region.	Cannot view unrelated regions.
AICTE System Administrator	Manage user/role assignments and approved policy configuration.	Does not automatically gain authority to certify institutional batches.
Degree-Awarding Body	Use permitted certificate/credential issue workflow.	Only authorised student records and issuer scope.
Verifier	Request/check permitted student information.	No access to private record before consent or a separately approved legal basis.
Student	View own record, consent to sharing, see access history and credential status.	Cannot directly rewrite official enrolment records.

4. Primary end-to-end user story

1. A College Operator logs in and downloads the current demo CSV template.

2. The operator uploads a synthetic CSV. The system creates an UploadBatch and stores rows as staged data.

3. Server-side validation records row number, rule code, severity, message and resolution status.

4. The operator corrects allowed fields and requests revalidation of changed rows.

5. The batch cannot be certified while blocking errors remain. Warnings are handled according to explicit policy.

6. The College Head reviews the declaration and certifies the submission.

7. The server atomically applies seat reservations and writes the official records, history and audit events, or rolls the operation back.

8. The dashboards show updated counts; conflict detection and risk-signal processing can follow.

9. A student can approve a mock verification request; a verifier sees only permitted fields. A public QR page shows minimal status.

5. Scope

MVP / must-have	Later / explicitly deferred
Login/logout, role-based authorization, institution/region scope.	MFA, enterprise federation and production identity-provider integration.
CSV upload, staging, seven validation-layer framework, error grid and revalidation.	High-volume durable queue and production object-storage infrastructure.
Configurable seat ledger, certification, commit, enrolment history and conflict case workflow.	Official quota automation until exact rules and approvals are verified.
College and AICTE dashboards backed by persisted demo data.	Advanced analytics warehouse, national-scale performance guarantees.
Rule-based anomaly signals with reasons and evidence.	Isolation Forest or any separate ML service.
Mock consent flow, QR status and audit-chain integrity demo.	Real APAAR/ABC/DigiLocker connections, production-grade signed credentials, HSM and legal integrations.

6. Explicit non-goals and prohibited claims

- The prototype is not an official AICTE system and has no implied approval or government affiliation.
- No real APAAR, AICTE EoA, ABC or DigiLocker API availability is assumed.
- No actual student data or real sensitive identifiers are required for demonstration.
- No quota percentage, reservation rule, age rule or current law is to be asserted from this specification alone; policy owners must confirm the official source and effective academic year.
- A flagged pattern does not prove fraud. Human review and evidence are required.
- A QR that contains a URL or identifier is not cryptographic proof of authenticity.

7. Demo acceptance storyline

Demo	Expected visible result
Upload a synthetic student file	Batch ID/status, row count and progress/status field appear.
Inject invalid identifier and duplicate	Exact row errors appear and block certification until resolved.
Show configured seat capacity	Valid categories pass; capacity overflow is rejected safely.
Show cross-college active enrolment	Conflict case is created; records are not labelled fraudulent automatically.
Open AICTE console	Aggregate KPIs and role-appropriate risk queue display persisted data.
Request student verification	Student approves; verifier receives only approved fields.
Revoke demo credential	Same QR/status endpoint reports revoked.
Alter an audit event in test fixture	Integrity verification detects mismatch and reports affected sequence.

8. Product success criteria versus measured results

The original report includes proposed performance targets and an estimated scale. Treat these as design targets, not existing results. Any claim such as “100,000 rows validated in under five minutes,” “99.9% uptime,” or “85% detection” must be labelled a target until measured through reproducible tests. Do not invent measured impact figures.

9. Open decisions

- Official academic-year seat capacities and category rules: TBD from valid approvals/policy owner.
- Exact student identifier format and external validation contract: TBD; demo format checks are synthetic-only.
- Whether verifier access can ever proceed without student consent: TBD with authorised legal/product owner; default prototype policy is consent required.
- Exact deployment provider and supported MongoDB topology: TBD.
- Formal signed credential format, issuer key management and revocation design: not part of first MERN MVP.